

一、[参考解析] (1) 利用 $f(x)$ 的一阶 Taylor 公式, 得

$$\lim_{n \rightarrow \infty} \left(\frac{f\left(a + \frac{1}{n}\right)}{f(a)} \right)^n = \lim_{n \rightarrow \infty} \left(1 + \frac{f'(a)}{f(a)} \cdot \frac{1}{n} + \frac{1}{f(a)} \cdot o\left(\frac{1}{n}\right) \right)^n = e^{\frac{f'(a)}{f(a)}}.$$

(2) 因为

$$I = \lim_{x \rightarrow 0} \frac{f(\sin^2 x + \cos x) \cdot 3x}{x^2 \cdot x} = 3 \lim_{x \rightarrow 0} \frac{f(\sin^2 x + \cos x)}{x^2},$$

所以

$$I = 3 \lim_{x \rightarrow 0} \frac{f(\sin^2 x + \cos x) - f(1)}{(\sin^2 x + \cos x) - 1} \cdot \frac{\sin^2 x + \cos x - 1}{x^2} = 3f'(1) \left(1 - \frac{1}{2}\right) = \frac{3}{2}f'(1).$$

(3) 由题设得

$$\frac{\partial z}{\partial x} = f'(e^x y^2) e^x y^2 = f(e^x y^2).$$

令 $u = e^x y^2$, 得到 $u > 0$, 并有 $f'(u)u = f(u)$, 即 $\frac{f'(u)}{f(u)} = \frac{1}{u}$, 从而

$$(\ln f(u))' = (\ln u)'.$$

所以有

$$\ln f(u) = \ln u + \ln C,$$

即

$$f(u) = Cu.$$

由初始条件 $f(1) = 2$ 解得 $C = 2$, 所以 $f(u) = 2u$. 故当 $x > 0$ 时, $f(x) = 2x$.

(4) 利用 Taylor 公式, 得

$$f(x) = \left[1 + x + \frac{1}{2!}x^2 + \frac{1}{3!}x^3 + o(x^3) \right] \cdot \left[2x - \frac{1}{3!}(2x)^3 + o(x^4) \right],$$

所以 $f(x)$ 展开式的 4 次项为

$$-\frac{1}{3!}(2x)^3 \cdot x + \frac{2}{3!}x^4 = -x^4,$$

从而 $\frac{f^{(4)}(0)}{4!} = -1$, 故 $f^{(4)}(0) = -24$.

(5) 该曲面在点 (x_0, y_0, z_0) 的切平面的法向量为 $(x_0, 2y_0, -1)$. 又该切平面与已知平面平行, 从而两平面法向量平行, 故

$$\frac{x_0}{2} = \frac{2y_0}{2} = \frac{-1}{-1},$$

从而 $x_0 = 2, y_0 = 1$, 得 $z_0 = \frac{x_0^2}{2} + y_0^2 = 3$, 从而所求切平面为

$$2(x-2) + 2(y-1) - (z-3) = 0,$$

即

$$2x + 2y - z = 3.$$

二、[参考证明] 由于 $f(0) = 0, f'(x) > 0$, 故 $f(x) > 0$. 记

$$F(x) = \left(\int_0^x f(t) dt \right)^2 - \int_0^x f^3(t) dx,$$

则 $F(0) = 0$, 且 $F'(x) = f(x)g(x)$, 其中

$$g(x) = 2 \int_0^x f(t) dt - f^2(x), \quad g(0) = 0.$$

当 $x > 0$ 时,

$$g'(x) = 2f(x)[1 - f'(x)] > 0,$$

从而 $g(x) > 0$. 进一步, $F'(x) > 0$, 因此当 $a > 0$ 时, $F(a) > F(0) = 0$, 不等式得证.

三、[参考解析] 由于 $\Omega: \left(x - \frac{1}{2}\right)^2 + \left(y - \frac{1}{2}\right)^2 + 2\left(z - \frac{1}{2}\right)^2 \leq 1$ 是一个椭球, 其体积为

$$V = \frac{2\sqrt{2}}{3}\pi.$$

作变换 $u = x - \frac{1}{2}, v = y - \frac{1}{2}, w = \sqrt{2}\left(z - \frac{1}{2}\right)$, 将 Ω 变为单位球 $\Sigma: u^2 + v^2 + w^2 \leq 1$,

而

$$\frac{\partial(u, v, w)}{\partial(x, y, z)} = \sqrt{2},$$

故 $du dv dw = \sqrt{2} dx dy dz$, 且

$$M = \frac{1}{\sqrt{2}} \iiint_{\Sigma} \left[\left(u + \frac{1}{2}\right)^2 + \left(v + \frac{1}{2}\right)^2 + \left(\frac{w}{\sqrt{2}} + \frac{1}{2}\right)^2 \right] du dv dw.$$

根据对称性, 一次项的积分都等于 0, 所以

$$M = \frac{1}{\sqrt{2}} \iiint_{\Sigma} \left[u^2 + v^2 + \frac{w^2}{2} \right] du dv dw + A,$$

其中 $A = \frac{1}{\sqrt{2}} \left(\frac{1}{4} + \frac{1}{4} + \frac{1}{4} \right) V = \frac{\pi}{2}$. 因为

$$I = \iiint_{\Sigma} (u^2 + v^2 + w^2) du dv dw = \int_0^{2\pi} d\varphi \int_0^{\pi} d\theta \int_0^1 \rho^2 \cdot \rho^2 \sin \theta d\rho = \frac{4\pi}{5},$$

仍由对称性, $\iiint_{\Sigma} u^2 du dv dw = \iiint_{\Sigma} v^2 du dv dw = \iiint_{\Sigma} w^2 du dv dw = \frac{I}{3}$, 所以

$$M = \frac{1}{\sqrt{2}} \left(\frac{1}{3} + \frac{1}{3} + \frac{1}{6} \right) I + A = \frac{2\sqrt{2}+3}{6} \pi.$$

四、[参考解析] [方法一] 将区间 $[0,1]$ 分成 n 等份, 设分点 $x_k = \frac{k}{n}$, 则 $\Delta x_k = \frac{1}{n}$, 且

$$\begin{aligned} & \lim_{n \rightarrow \infty} n \left(\int_0^1 f(x) dx - \frac{1}{n} \sum_{k=1}^n f\left(\frac{k}{n}\right) \right) = \lim_{n \rightarrow \infty} n \left(\sum_{k=1}^n \int_{x_{k-1}}^{x_k} f(x) dx - \sum_{k=1}^n f(x_k) \Delta x_k \right) \\ &= \lim_{n \rightarrow \infty} n \left(\sum_{k=1}^n \int_{x_{k-1}}^{x_k} [f(x) - f(x_k)] dx \right) = \lim_{n \rightarrow \infty} n \left(\sum_{k=1}^n \int_{x_{k-1}}^{x_k} \frac{f(x) - f(x_k)}{x - x_k} (x - x_k) dx \right) \\ &= \lim_{n \rightarrow \infty} n \left(\sum_{k=1}^n \frac{f(\xi_k) - f(x_k)}{\xi_k - x_k} \int_{x_{k-1}}^{x_k} (x - x_k) dx \right) \quad (\text{其中 } \xi_k \in (x_{k-1}, x_k)) \\ &= \lim_{n \rightarrow \infty} n \left(\sum_{k=1}^n f'(\eta_k) \int_{x_{k-1}}^{x_k} (x - x_k) dx \right) \quad (\text{其中 } \eta_k \text{ 在 } \xi_k, x_k \text{ 之间}) \\ &= \lim_{n \rightarrow \infty} n \left(\sum_{k=1}^n f'(\eta_k) \left(-\frac{1}{2} (x_k - x_{k-1})^2 \right) \right) \\ &= \lim_{n \rightarrow \infty} \frac{-1}{2} \left(\sum_{k=1}^n f'(\eta_k) (x_k - x_{k-1}) \right) = -\frac{1}{2} \int_0^1 f'(x) dx = -\frac{1}{2}. \end{aligned}$$

[方法二] 记 $F(x) = \int_0^x f(t) dt$, 则 $F(0) = 0$, $F'(x) = f(x)$. 因此

$$\int_0^1 f(x) dx = F(1) - F(0) = \sum_{k=1}^n \left[F\left(\frac{k}{n}\right) - F\left(\frac{k-1}{n}\right) \right],$$

利用 Taylor 公式, 存在 $\xi_k \in \left(\frac{k-1}{n}, \frac{k}{n}\right)$, 使得

$$\begin{aligned} F\left(\frac{k-1}{n}\right) &= F\left(\frac{k}{n}\right) - \frac{1}{n}F'\left(\frac{k}{n}\right) + \frac{1}{2n^2}F''(\xi_k), \\ F\left(\frac{k}{n}\right) - F\left(\frac{k-1}{n}\right) - \frac{1}{n}f\left(\frac{k}{n}\right) &= -\frac{1}{2n^2}f'(\xi_k), \end{aligned}$$

于是, 得

$$\begin{aligned} \lim_{n \rightarrow \infty} n \left(\int_0^1 f(x) dx - \frac{1}{n} \sum_{k=1}^n f\left(\frac{k}{n}\right) \right) &= -\frac{1}{2} \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{k=1}^n f'(\xi_k) = -\frac{1}{2} \int_0^1 f'(x) dx \\ &= -\frac{1}{2}(f(1) - f(0)) = -\frac{1}{2}. \end{aligned}$$

五、[参考证明] 设 $F(x) = \frac{1}{I} \int_0^x f(t) dt$, 则 $F(0) = 0, F(1) = 1$, 且 $F'(x) = \frac{f(x)}{I}$.

根据连续函数的介值定理, 存在 $\xi \in (0, 1)$, 使得 $F(\xi) = \frac{1}{2}$.

分别在两个子区间 $[0, \xi], [\xi, 1]$ 上应用 Lagrange 中值定理, 得

$$\begin{aligned} F'(x_1) &= \frac{f(x_1)}{I} = \frac{F(\xi) - F(0)}{\xi - 0} = \frac{1/2}{\xi}, \quad x_1 \in (0, \xi), \\ F'(x_2) &= \frac{f(x_2)}{I} = \frac{F(1) - F(\xi)}{1 - \xi} = \frac{1/2}{1 - \xi}, \quad x_2 \in (\xi, 1), \end{aligned}$$

所以

$$\frac{1}{f(x_1)} + \frac{1}{f(x_2)} = \frac{1}{I} \left[\frac{1}{F'(x_1)} + \frac{1}{F'(x_2)} \right] = \frac{1}{I} \left[\frac{\xi}{1/2} + \frac{1-\xi}{1/2} \right] = \frac{2}{I}.$$

六、[参考证明] 由 $f(x) = f(x+2)$ 知, $f(x)$ 为以 2 为周期的周期函数, 其 Fourier 系数为

$$a_n = \int_{-1}^1 f(x) \cos n\pi x dx, \quad b_n = \int_{-1}^1 f(x) \sin n\pi x dx,$$

再由 $f(x) = f(x + \sqrt{3})$ 得

$$\begin{aligned} a_n &= \int_{-1}^1 f(x + \sqrt{3}) \cos n\pi x dx \\ &= \int_{-1+\sqrt{3}}^{1+\sqrt{3}} f(t) \cos n\pi(t - \sqrt{3}) dt \\ &= \int_{-1+\sqrt{3}}^{1+\sqrt{3}} f(t) (\cos n\pi t \cos \sqrt{3}n\pi + \sin n\pi t \sin \sqrt{3}n\pi) dt \end{aligned}$$

$$\begin{aligned}
&= \cos \sqrt{3}n\pi \int_{-1+\sqrt{3}}^{1+\sqrt{3}} f(t) \cos n\pi t dt + \sin \sqrt{3}n\pi \int_{-1+\sqrt{3}}^{1+\sqrt{3}} f(t) \sin n\pi t dt \\
&= \cos \sqrt{3}n\pi \int_{-1}^1 f(t) \cos n\pi t dt + \sin \sqrt{3}n\pi \int_{-1}^1 f(t) \sin n\pi t dt,
\end{aligned}$$

所以 $a_n = a_n \cos \sqrt{3}n\pi + b_n \sin \sqrt{3}n\pi$.

同理可得 $b_n = b_n \cos \sqrt{3}n\pi - a_n \sin \sqrt{3}n\pi$.

联立解得 $a_n = b_n = 0$ ($n = 1, 2, \dots$).

因为 $f(x)$ 在 $(-\infty, +\infty)$ 上处处可导, 其 Fourier 级数处处收敛于 $f(x)$, 所以

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx) = \frac{a_0}{2},$$

其中 $a_0 = \int_{-1}^1 f(x) dx$ 为常数.